

Day 2 — TNPSC Group 2A Study Material

Part A – General Studies: Nature of the Universe & General Scientific Laws

READING MATERIAL

Two adjoining one-line syllabus clauses — 'Nature of universe' and 'General scientific laws' — bundled into one topic since neither alone has enough distinct testable content for a full day. Universe questions tend to be single-fact recall (orbital periods, terminology); scientific-law questions tend to test everyday-life applications of Newton's laws, which is exactly how the syllabus itself frames this unit ('everyday application of basic principles of mechanics').

Q: How long does Earth take to orbit the Sun, and how long does Mars take?

A: Earth: 365 days. Mars: 687 days (about 1.88 Earth years, since Mars is farther out and moves slower).

Q: What is a galaxy, and what is the name of ours?

A: A galaxy is a system of billions of stars (along with gas, dust, and dark matter) bound together by gravity. Our galaxy is the Milky Way.

Q: What is the Big Bang?

A: The event when all matter was confined to a single point and began to expand outward — the leading scientific model for the origin of the universe.

Q: How did the universe's temperature change in the minutes and millennia after the Big Bang?

A: Within about 3 minutes, it had dropped below roughly 1 billion degrees. By about 300,000 years after the Big Bang, it had cooled further to around 3,000 degrees (the 'recombination' era, when the universe became transparent — note that cosmologists usually express this in Kelvin rather than Celsius, so treat the exact number cautiously, but '~3,000' is the figure these prep sources expect).

Q: What gases filled the early universe at that cooling stage?

A: Clouds of hydrogen and helium gas — the two simplest, lightest elements, which is why they were the first to form after the Big Bang.

Q: What is the only direct evidence of the Big Bang itself?

A: The Cosmic Microwave Background (CMB) — a faint glow of radiation filling all of space, left over from the early hot universe.

Q: What is an Astronomical Unit (AU)?

A: The average distance between the Earth and the Sun, denoted 'au'. $1 \text{ au} = 1.496 \times 10^8 \text{ km}$ (about 149.6 million km).

Q: The law of conservation of angular momentum is considered a consequence of which of Newton's laws?

A: Newton's Third Law — internal action-reaction force pairs within a system produce equal and opposite torques that cancel out, so the system's total angular momentum stays conserved when no external torque acts.

Q: In which everyday sport is the 'turning effect of force' (torque) directly used?

A: Cycling — pedalling applies a turning force (torque) to the pedal-crank, which is a textbook example of torque in daily life.

Q: Why do passengers lean forward when a moving vehicle brakes suddenly?

A: Inertia of motion — a body in motion tends to continue moving in the same direction/speed unless an external force acts on it; the passenger's body keeps moving forward even as the vehicle decelerates.

Q: Which theory explains the behaviour of matter and energy in black holes?

A: The Theory of Relativity (Einstein's General Relativity, specifically) — it describes how gravity warps spacetime so extremely that nothing, not even light, escapes a black hole.

Q: What physical quantity is defined as the product of a moving body's mass and velocity?

A: Linear momentum ($p = mv$).

Q: What everyday quantity directly changes a car's speed?

A: A net force causes acceleration, and it's that acceleration which changes the car's speed (Newton's Second Law, $F=ma$). Some prep materials phrase the question as simply 'what changes the car's speed' and mark 'acceleration' as the answer rather than 'force' — worth knowing both framings since the exact expected answer depends on how the question is worded.

MOCK TEST (WITH ANSWERS)

1. Earth completes one orbit around the Sun in approximately:

- A. 180 days
- B. 365 days ✓**
- C. 687 days
- D. 30 days

Explanation: 365 days (one year). [easy, web-research]

2. Mars takes approximately how many days to orbit the Sun?

- A. 250 days
- B. 365 days
- C. 687 days ✓**
- D. 853 days

Explanation: About 687 Earth days. [easy, web-research]

3. What is the name of our home galaxy?

- A. Andromeda Galaxy
- B. Milky Way Galaxy ✓**
- C. Black Eye Galaxy
- D. Bode's Galaxy

Explanation: The Milky Way. [easy, web-research]

4. The event when matter confined to a single point began to expand, giving rise to the universe, is called:

- A. A black hole
- B. A galaxy formation
- C. The Big Bang ✓**
- D. A supernova

Explanation: The Big Bang. [easy, web-research]

5. The only direct observational evidence of the Big Bang is:

- A. A cluster of stars
- B. The Hubble telescope
- C. A black hole

D. Cosmic Microwave Background radiation ✓

Explanation: Cosmic Microwave Background (CMB). [medium, web-research]

6. The average distance between Earth and the Sun is called:

- A. A light year
- B. A parsec
- C. An Astronomical Unit ✓**
- D. A solar radius

Explanation: Astronomical Unit (au), $\approx 1.496 \times 10^8$ km. [easy, web-research]

7. 1 Astronomical Unit is approximately equal to:

- A. 1.496×10^8 km ✓**
- B. 2.985×10^8 km
- C. 1.893×10^8 km
- D. 7.418×10^8 km

Explanation: 1 au $\approx 1.496 \times 10^8$ km. [medium, web-research]

8. The law of conservation of angular momentum is a consequence of:

- A. Newton's First Law
- B. Newton's Second Law
- C. Newton's Third Law ✓**
- D. The Law of Gravitation

Explanation: Internal action-reaction torque pairs (Newton's Third Law) cancel, conserving total angular momentum. [hard, web-research]

9. A passenger leans forward when a bus brakes suddenly. This is best explained by:

- A. Inertia of rest

B. Inertia of motion ✓

C. Inertia of direction

D. Centripetal force

Explanation: Inertia of motion — the body continues moving forward as the vehicle decelerates. [easy, web-research]

10. The Theory of Relativity is most directly used to explain the behaviour of matter and energy in:

A. Volcanoes

B. Black holes ✓

C. Ocean currents

D. Chemical reactions

Explanation: General Relativity describes extreme gravity, as found around black holes. [medium, web-research]

11. Linear momentum of a moving body is defined as the product of its:

A. Mass and acceleration

B. Mass and velocity ✓

C. Force and time

D. Velocity and time

Explanation: Momentum $p = \text{mass} \times \text{velocity}$. [easy, web-research]

12. The early universe, shortly after the Big Bang, was filled mainly with clouds of which gases?

A. Oxygen and Nitrogen

B. Hydrogen and Helium ✓

C. Carbon dioxide and Methane

D. Helium and Neon

Explanation: Hydrogen and Helium — the two lightest, simplest elements formed first. [medium, web-research]

Part B – Aptitude & Mental Ability: Percentage

READING MATERIAL

Percentage is 1 of 10 named Aptitude sub-skills. TNPSC's percentage questions are built from a handful of recurring shapes — successive discounts, reverse percentage ('20% of $x = 50$, find x '), percentage of a mixture, and percentage of a time period — recognize the shape first, then compute.

Q: Method — successive percentage discounts: A TV was sold for ₹14,400 after successive discounts of 10% and 20%. What was its original (market) price?

A: Successive discounts don't add (it's NOT a flat 30% off) — apply them one after another. If the market price is P : $P \times 0.90 \times 0.80 = 14,400 \rightarrow P \times 0.72 = 14,400 \rightarrow P = 14,400 / 0.72 = ₹20,000$.

Q: Method — percentage of a mixture (the parts must add to 100%): An alloy is 30% copper and 40% zinc, rest nickel. Find the nickel in 20 kg of the alloy.

A: Nickel % = $100 - 30 - 40 = 30\%$. Nickel amount = 30% of 20 kg = 6 kg. The key step is realizing the percentages must total 100% — nickel is whatever's left over, not a number given directly.

Q: Method — reverse percentage ('find the whole, given a percentage of it'): If 20% of $x = 50$, find x .

A: 20% of $x = 50 \rightarrow 0.20 \times x = 50 \rightarrow x = 50 / 0.20 = 250$. Reverse-percentage questions give you the part and the percentage, and ask for the whole — divide, don't multiply.

Q: Method — percentage of a decimal/small number: Find 0.75% of 40 kg.

A: $0.75\% = 0.0075$. $0.0075 \times 40 = 0.3$ kg. Common mistake: treating '0.75%' as '0.75' (i.e. 75%) instead of converting the decimal percentage properly — always convert to a decimal fraction first (divide by 100) before multiplying.

Q: Method — percentage of a time period: What percentage of a day is 10 hours?

A: A day = 24 hours. $10/24 \times 100 = 41.67\%$.

Q: Method — percentage of a fraction: What percent of $2/7$ is $1/35$?

A: Set up: $x\% \text{ of } (2/7) = 1/35 \rightarrow x/100 \times 2/7 = 1/35 \rightarrow x = (1/35) \times (7/2) \times 100 = (7/70) \times 100 = 10\%$.

Q: Method — sum of two independent percentage-of-number terms: Find the value of 28% of 450 + 45% of 280.

A: 28% of 450 = 126. 45% of 280 = 126. Sum = $126 + 126 = 252$. (A nice coincidence both terms equal 126 here — don't expect that in general, just add the two percentage values separately.)

Q: Method — percentage greater than 100% (the result is bigger than the original number): Find 280% of 3,940.

A: $280\% = 2.80$. $2.80 \times 3,940 = 11,032$. Percentages over 100% just mean the result is larger than the base number — the calculation method doesn't change.

MOCK TEST (WITH ANSWERS)

1. A mobile phone was sold for ₹10,200 after successive discounts of 15% and 20%. What was its original price?

- A. ₹14,500
- B. ₹15,000 ✓**
- C. ₹15,500
- D. ₹16,000

Explanation: $P \times 0.85 \times 0.80 = 10,200 \rightarrow P \times 0.68 = 10,200 \rightarrow P = 15,000$. [medium, authored]

2. An alloy is 25% tin and 35% lead, the rest is copper. Find the copper in 40 kg of the alloy.

- A. 10 kg
- B. 14 kg
- C. 16 kg ✓**
- D. 20 kg

Explanation: Copper % = $100 - 25 - 35 = 40\%$. 40% of 40 kg = 16 kg. [easy, authored]

3. If 35% of $y = 91$, find y .

- A. 230
- B. 250
- C. 260 ✓**

D. 240

Explanation: $y = 91 / 0.35 = 260$. [easy, authored]

4. Find 1.25% of 80 kg.

A. 1 kg ✓

B. 10 kg

C. 0.1 kg

D. 12.5 kg

Explanation: $1.25\% = 0.0125$. $0.0125 \times 80 = 1$ kg. [easy, authored]

5. What percentage of a day is 6 hours?

A. 20%

B. 25% ✓

C. 30%

D. 16.67%

Explanation: $6/24 \times 100 = 25\%$. [easy, authored]

6. What percent of $3/8$ is $3/40$?

A. 10%

B. 15%

C. 20% ✓

D. 12%

Explanation: $x\%$ of $(3/8) = 3/40$, so $x/100 = (3/40) \div (3/8) = (3/40) \times (8/3) = 1/5$, giving $x = 20\%$. [medium, authored]

7. Find the value of 32% of 350 + 18% of 200.

A. 148 ✓

B. 138

C. 150

D. 146

Explanation: 32% of 350 = 112. 18% of 200 = 36. Sum = 148. [medium, authored]

8. Find 320% of 2,500.

A. 7,500

B. 8,000 ✓

C. 8,500

D. 9,000

Explanation: $320\% = 3.2$. $3.2 \times 2,500 = 8,000$. [easy, authored]

9. A quantity is first increased by 20%, then decreased by 20%. What is the net percentage change?

A. No change

B. 4% decrease ✓

C. 4% increase

D. 20% decrease

Explanation: Net change after +20% then -20% is always a decrease of $(20 \times 20)/100 = 4\%$ — increasing and then decreasing by the same percentage never cancels out, because the second change applies to a different (larger) base. [hard, authored]

10. The price of an item increases from ₹250 to ₹300. What is the percentage increase?

A. 15%

B. 20% ✓

C. 25%

D. 30%

Explanation: Increase = 50. Percentage increase = $(50/250) \times 100 = 20\%$. [easy, authored]

Part C – General English: Poems: "The Crocodile" (Lewis Carroll) & "From a Railway Carriage" (R.L. Stevenson)

READING MATERIAL

Two short poems bundled into one day. Both are tested almost entirely through figure-of-speech identification on specific lines (matching the syllabus's explicit framing for Unit VII), so the Q&A below pairs each notable line with its device, plus the background/context that 'appreciation' questions tend to probe.

Q: Where does "The Crocodile" actually come from, and is it an original poem?

A: It's a parody. Lewis Carroll wrote it as a parody of Isaac Watts' moralistic poem 'Against Idleness and Mischief' ('How doth the little busy bee...'). Carroll's version appears as a poem recited by Alice in Chapter 2 of Alice's Adventures in Wonderland, originally titled 'How Doth the Little Crocodile'.

Q: What is the crocodile doing with its tail, and what mood does it project?

A: It is improving/grooming its shining tail to make it shine more. The crocodile is portrayed as cunning and outwardly cheerful.

Q: Identify the figure of speech: "How cheerful he seems to grin."

A: Personification (and the line is also cited as an example of alliteration in some sources — 'cheerful' / 'seems' sound patterns) — attributing a human-like cheerful expression to the crocodile.

Q: Identify the figure of speech: "Golden scale" describing the crocodile's skin.

A: Metaphor — the scale isn't literally gold, it's being described AS gold for its gleam.

Q: What do the 'golden scales' suggest about the crocodile, beyond just appearance?

A: An attractive trap — the glittering, attractive appearance lures fish in, even though the crocodile's true intent is predatory. This is the poem's central irony.

Q: Identify the figure of speech: "And welcomes little fishes in... with gently smiling jaws."

A: Irony — the crocodile 'welcomes' the fish in a friendly-sounding way, but the true purpose is to eat them. The gap between the gentle/welcoming language and the predatory reality is the poem's comic-dark irony. ('Smiling jaws' on its own is also cited as personification — a gentle smiling expression is a human trait, here applied to an animal's mouth.)

Q: Why does the crocodile grin, according to the poem's logic?

A: To deceive the fish — the cheerful, welcoming appearance is a lure, not genuine friendliness.

Q: Why does the poet mention the River Nile?

A: To establish the crocodile's natural habitat — grounding the otherwise whimsical, talking-animal poem in a real geographical setting.

Q: What is the overall lesson/moral the poet conveys through "The Crocodile"?

A: "Be careful of appearances" — cheerful, welcoming surfaces can hide dangerous intentions, a theme reinforced by the poem itself being a parody of a much more sincere moral poem (Watts' bee poem about industriousness).

Q: What is the setting and origin of "From a Railway Carriage"?

A: It's narrated from a child's point of view, looking out of a train window at the rapidly passing countryside. It was published in Robert Louis Stevenson's 1885 collection A Child's Garden of Verses. The poem's galloping rhythm is written to mimic the rhythmic chugging/motion of a moving train.

Q: Identify the figure of speech: "Faster than fairies, faster than witches."

A: Simile (uses 'than' to compare the train's speed to fairies and witches) — and the same line is also cited as alliteration, for the repeated 'f'/'w' sounds across 'faster'/'fairies' and the repeated structure.

Q: Identify the figure of speech: "Bridges and houses, hedges and ditches."

A: Alliteration — paired repeated consonant sounds ('h' in houses/hedges, and the rhythmic pairing itself) that mimic the train's clattering rhythm.

Q: Identify the figure of speech: "Charging along like troops in a battle" and "Fly as thick as driving rain."

A: Both are similes — direct comparisons using 'like'/'as', describing how things outside the train rush past.

Q: Identify the figure of speech: "And ever again, in the wink of an eye."

A: Metaphor (an idiom used metaphorically) — 'wink of an eye' isn't literal, it stands for an extremely brief moment.

Q: Identify the figure of speech: "Painted stations whistle by."

A: This line does double duty: Personification (stations don't literally 'whistle' — a human/animal-like action is given to an inanimate object) AND Onomatopoeia ('whistle' is itself a sound-word). Different MCQs may ask for either device on this exact line.

Q: Identify the figure of speech: "Here is a child who clammers and scrambles" and "Lumping along with man and load."

A: Both are cited as consonance — repetition of matching consonant sounds at the end of nearby words (clammers/scrambles; lumping/man/load), distinct from alliteration which repeats STARTING sounds.

MOCK TEST (WITH ANSWERS)

1. "The Crocodile" by Lewis Carroll is a parody of which earlier poem?

A. Wordsworth's 'Daffodils'

B. Isaac Watts' 'Against Idleness and Mischief' ✓

C. Blake's 'The Tyger'

D. Stevenson's 'Happy Thought'

Explanation: It parodies Isaac Watts' moralistic 'Against Idleness and Mischief' ('How doth the little busy bee'). [hard, authored]

2. In which work does Lewis Carroll's crocodile poem originally appear?

A. Through the Looking-Glass

B. Alice's Adventures in Wonderland ✓

C. The Hunting of the Snark

D. Sylvie and Bruno

Explanation: Alice recites it in Chapter 2 of Alice's Adventures in Wonderland. [hard, authored]

3. "Golden scale" describing the crocodile's skin is an example of:

A. Simile

B. Metaphor ✓

C. Onomatopoeia

D. Hyperbole

Explanation: Metaphor — directly calling the scale 'golden' rather than comparing with 'like/as'. [medium, web-research]

4. The crocodile 'welcoming' little fishes in, only to eat them, is best described as an example of:

A. Simile

B. Hyperbole

C. Irony ✓

D. Euphemism

Explanation: Irony — the gap between the friendly 'welcome' and the predatory reality. [medium, web-research]

5. What overall lesson does "The Crocodile" convey?

A. Always swim in groups

B. Be careful of appearances ✓

C. Fish are smarter than crocodiles

D. Rivers are dangerous

Explanation: A cheerful, welcoming surface can hide dangerous intent. [easy, web-research]

6. "From a Railway Carriage" was published in which Stevenson collection?

A. Treasure Island

B. A Child's Garden of Verses ✓

C. Kidnapped

D. The Black Arrow

Explanation: Published in A Child's Garden of Verses (1885). [hard, authored]

7. "Faster than fairies, faster than witches" is an example of:

A. Personification

B. Simile ✓

C. Onomatopoeia

D. Oxymoron

Explanation: A simile, comparing the train's speed using 'than'. [easy, web-research]

8. "Charging along like troops in a battle" uses which figure of speech?

A. Metaphor

B. Simile ✓

C. Personification

D. Hyperbole

Explanation: Simile — direct comparison using 'like'. [easy, web-research]

9. "Painted stations whistle by" — which figure of speech gives stations a human/animal-like action?

A. Onomatopoeia

B. Personification ✓

C. Simile

D. Alliteration

Explanation: Personification — stations can't literally whistle. [medium, web-research]

10. In the same line, "whistle by" is also an example of which device, since 'whistle' imitates a sound?

A. Onomatopoeia ✓

B. Metaphor

C. Consonance

D. Hyperbole

Explanation: Onomatopoeia — 'whistle' is a sound-imitating word. [medium, web-research]

11. "Here is a child who clambers and scrambles" exhibits which sound device?

A. Alliteration

B. Consonance ✓

C. Assonance

D. Onomatopoeia

Explanation: Consonance — matching consonant sounds at the end of nearby words. [medium, web-research]

12. "And ever again, in the wink of an eye" uses 'wink of an eye' as a:

A. Simile

B. Metaphor ✓

C. Personification

D. Onomatopoeia

Explanation: Metaphor — a brief-moment idiom used non-literally. [medium, web-research]